

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

Course No: CSE 2104

Course Title: Data Structures Lab

Assignment No: 01

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Lab Section: A2

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Program: BSc in CSE

Question :

A bookstore maintains a sorted list of unique book IDs to quickly identify whether a book is available in stock. You are required to write a C++ program that uses Binary Search to determine if a specific book ID is present in the list. Additionally, the store often needs to find the first book ID that is not smaller than a given value (lower bound) and the first book ID that is greater than a given value (upper bound) in order to manage restocking and categorization.

Given a sorted array of book IDs and a query book ID, your program should output whether the book exists in the list, the lower bound position and the upper bound position corresponding to the query.

Tasks

1. Read the number of book IDs and then read the sorted list of unique book IDs.
2. Read the query book ID to be searched.
3. Implement Binary Search to check if the book ID exists.
4. Implement logic to find the lower bound of the query.
5. Implement logic to find the upper bound of the query.
6. Display the search result along with the lower and upper bound indices.

You should write user defined functions for Binary Search, Lower Bound and Upper Bound.

Input should be taken inside the main function. All user defined functions must be called from

the main function.

Sample Input	Sample Output
7 10 20 30 40 50 60 70 40	Book Found Lower Bound Index: 3 Upper Bound Index: 4

Solution :

```
#include <bits/stdc++.h>
```

```
using namespace std;
```

```
int linearSearch(vector<int> arr, int tar, int n) {
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (arr[i] == tar) {
```

```
            return i;
```

```
        }
```

```
    }
```

```
    return -1;
```

```
}
```

```
int binarySearch(vector<int> arr, int tar, int n) {
```

```
    int low = 0;
```

```
    int high = n - 1;
```

```
    while (low <= high) {
```

```
        int mid = low + (high - low) / 2;
```

```
        if (arr[mid] == tar) {
```

```
            return mid;
```

```
        } else if (arr[mid] < tar) {
```

```
            low = mid + 1;
```

```
        } else {
```

```
            high = mid - 1;
```

```
        }
```

```

    }
    return -1;
}

int getLowerBound(vector<int> arr, int tar, int n) {
    int low = 0;
    int high = n - 1;

    int ans = n;

    while (low <= high) {
        int mid = low + (high - low) / 2;

        if (arr[mid] >= tar) {
            ans = mid;
            high = mid - 1;
        } else {
            low = mid + 1;
        }
    }
    return ans;
}

int getUpperBound(vector<int> arr, int tar, int n) {
    int low = 0;
    int high = n - 1;

```

```
int ans = n;

while (low <= high) {
    int mid = low + (high - low) / 2;

    if (arr[mid] <= tar) {
        low = mid + 1;
    } else {
        ans = mid;
        high = mid - 1;
    }
}

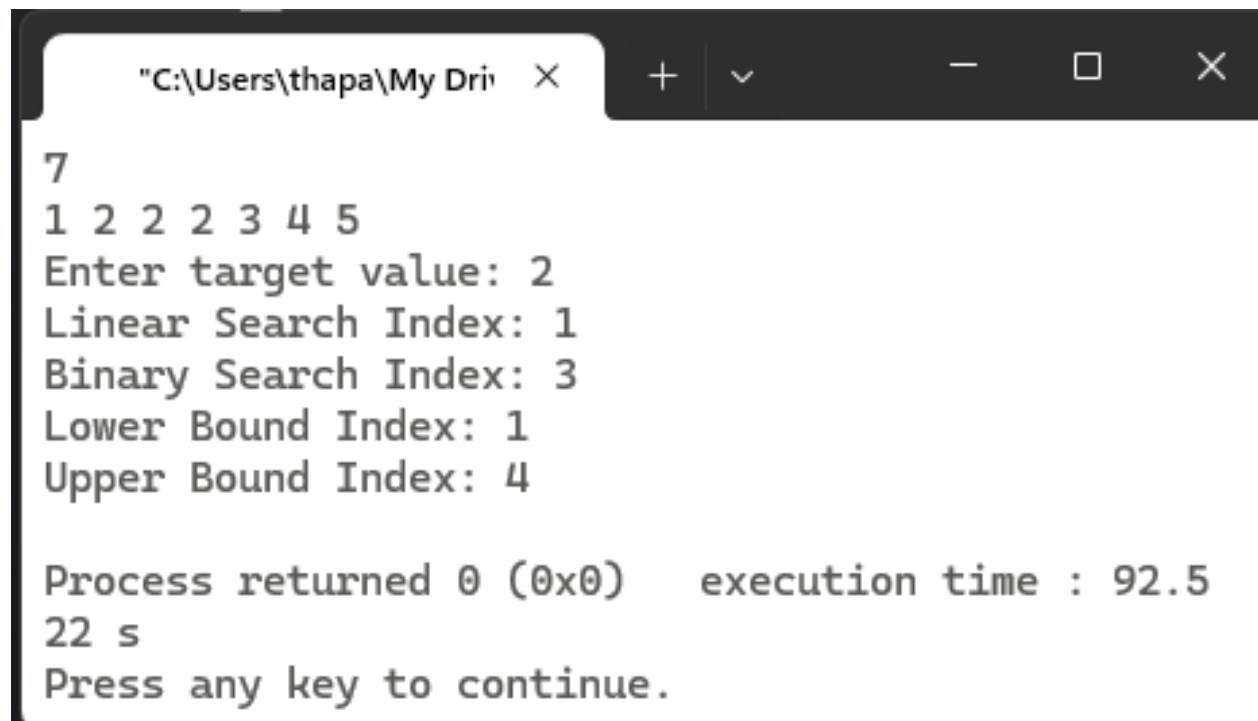
return ans;
}
```

```
int main(){
    int n,tar;
    cin >> n;
    vector<int> arr;

    for(int i=0; i < n; i++){
        int x;
        cin >> x;
        arr.push_back(x);
    }
```

```
cout << "Enter target value: ";  
cin >> tar;  
  
cout << "Linear Search Index: " << linearSearch(arr, tar, n) << endl;  
cout << "Binary Search Index: " << binarySearch(arr, tar, n) << endl;  
cout << "Lower Bound Index: " << getLowerBound(arr, tar, n) << endl;  
cout << "Upper Bound Index: " << getUpperBound(arr, tar, n) << endl;  
  
}
```

Output :



```
"C:\Users\thapa\My Drive" X + v - □ X  
7  
1 2 2 2 3 4 5  
Enter target value: 2  
Linear Search Index: 1  
Binary Search Index: 3  
Lower Bound Index: 1  
Upper Bound Index: 4  
  
Process returned 0 (0x0)    execution time : 92.5  
22 s  
Press any key to continue.
```